

HDOC DAY-21

1. Problem Statement

Write a menu-driven C program that accepts two positive integers, m and n , where $m \leq n$. Display an interactive menu using a switch statement to allow users to:

1. Print the first 10 multiples for integers from m to n .
2. Print addition tables for integers from m to n using addends 1 to 10.
3. Print subtraction tables for integers from m to n using subtrahends 1 to 10.
4. Print division results for integers from m to n divided by 1 to 10, rounded to two decimal places.
5. Exit the program.

Validate inputs to ensure $m > 0$, $n > 0$, and $m \leq n$.

Display suitable error messages for invalid menu choices and continue processing until the exit option is chosen.

2. Algorithms

Main Execution Flow

1. Prompt user for m and n .
2. Validate that $m > 0$, $n > 0$, and $m \leq n$; re-prompt if invalid.

3. Display main menu in a do-while loop.
4. Read user choice and execute corresponding function using switch-case.
5. Loop until choice equals 5.

Operation Algorithms

- **print_multiples(m, n):** Loop num from m to n . Inner loop i from 1 to 10: compute $\text{num} * i$ and print.
- **print_addition_table(m, n):** Loop num from m to n . Inner loop i from 1 to 10: compute $\text{num} + i$ and print.
- **print_subtraction_table(m, n):** Loop num from m to n . Inner loop i from 1 to 10: compute $\text{num} - i$ and print.
- **print_division_table(m, n):** Loop num from m to n . Inner loop i from 1 to 10: compute $(\text{float})\text{num} / i$ and print formatted to 2 decimal places ($\%.2f$).

3. Test Case Table

Test Case	Scenario	Input m, n	Menu Choice	Expected Output	Actual Output	Status
1	Negative Input	-2, 5	N/A	Error message &	Error message	Pass

Test Case	Scenario	Input m, n	Menu Choice	Expected Output	Actual Output	Status
				prompt again	displayed	
2	Invalid Range ($m > n$)	7, 3	N/A	Error message & prompt again	Error message displayed	Pass
3	Valid Range Input	2, 3	N/A	Main menu displayed	Main menu displayed	Pass
4	Multiplies Option	2, 3	1	Multiplies 1-10 for 2 and 3	Printed correctly	Pass

Tes t Ca se	Scenar io	Inp ut m, n	Men u Cho ice	Expect ed Output	Actual Output	Stat us
5	Additio n Option	2, 3	2	Additio n table for 2 and 3	Printe d correc tly	Pas s
6	Subtra ction Option	2, 3	3	Subtra ction table for 2 and 3	Printe d correc tly	Pas s
7	Divisio n Option	2, 3	4	Divisio n table with 2 decima ls	Printe d correc tly	Pas s
8	Invalid Choice	2, 3	9	Error: Invalid choice	Error messa ge	Pas s

Tes t Ca se	Scenar io	Inp ut m, n	Men u Cho ice	Expect ed Output	Actual Output	Stat us
				messa ge	displa yed	
9	Exit Option	2, 3	5	"Exiting progra m..."	Progra m termin ates	Pas s

4. Source Code Listing

```
#include <stdio.h>

void print_multiples(int m, int n) {
    for (int num = m; num <= n; num++) {
        printf("\n--- First 10 Multiples of %d ---\n", num);
        for (int i = 1; i <= 10; i++) {
            printf("%d x %2d = %d\n", num, i, num * i);
        }
    }
}

void print_addition_table(int m, int n) {
    for (int num = m; num <= n; num++) {
        printf("\n--- Addition Table for %d (Addends 1 to 10) ---\n", num);
        for (int i = 1; i <= 10; i++) {
            printf("%d + %2d = %d\n", num, i, num + i);
        }
    }
}
```

```

void print_subtraction_table(int m, int n) {
    for (int num = m; num <= n; num++) {
        printf("\n--- Subtraction Table for %d (Subtrahends 1 to 10) ---\n",
num);
        for (int i = 1; i <= 10; i++) {
            printf("%d - %2d = %d\n", num, i, num - i);
        }
    }
}

```

```

void print_division_table(int m, int n) {
    for (int num = m; num <= n; num++) {
        printf("\n--- Division Table for %d (Divisors 1 to 10) ---\n", num);
        for (int i = 1; i <= 10; i++) {
            printf("%d / %2d = %.2f\n", num, i, (float)num / i);
        }
    }
}

```

```

int main(void) {
    int m, n;
    int choice;

    do {
        printf("Enter first positive integer (m): ");
        scanf("%d", &m);
        printf("Enter second positive integer (n, where n >= m): ");
        scanf("%d", &n);

        if (m <= 0 || n <= 0) {
            printf("Error: Both m and n must be positive integers (> 0). Try
again.\n\n");
        } else if (m > n) {
            printf("Error: m must be less than or equal to n (m <= n). Try
again.\n\n");
        }
    } while (m <= 0 || n <= 0 || m > n);

    do {

```

```

printf("\n===== MENU (m = %d, n = %d)
=====\\n", m, n);
printf("1. Print first 10 multiples of each integer from m to n\\n");
printf("2. Print addition table of each integer from m to n (addends 1
to 10)\\n");
printf("3. Print subtraction table of each integer from m to n
(subtrahends 1 to 10)\\n");
printf("4. Print division table of each integer from m to n (divisors 1 to
10)\\n");
printf("5. Exit\\n");
printf("Enter your choice (1-5): ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        print_multiples(m, n);
        break;
    case 2:
        print_addition_table(m, n);
        break;
    case 3:
        print_subtraction_table(m, n);
        break;
    case 4:
        print_division_table(m, n);
        break;
    case 5:
        printf("Exiting program...\\n");
        break;
    default:
        printf("Error: Invalid menu choice! Please select an option
between 1 and 5.\\n");
        break;
}
} while (choice != 5);

return 0;
}

```

6. Execution & Verification Screenshots



```
kali@kali: ~/HDQC_Day21
Session Actions Edit View Help
(kali@kali)-[~]
$ mkdir HDQC_Day21
cd HDQC_Day21
(kali@kali)-[~/HDQC_Day21]
$ nano day21.c
(kali@kali)-[~/HDQC_Day21]
$ ls
day21.c
(kali@kali)-[~/HDQC_Day21]
$ gcc day21.c -o day21
(kali@kali)-[~/HDQC_Day21]
$ ./day21
Enter first positive integer (m): 2
Enter second positive integer (n, where n ≥ m): 3
===== MENU (m = 2, n = 3) =====
1. Print first 10 multiples of each integer from m to n
2. Print addition table of each integer from m to n (addends 1 to 10)
3. Print subtraction table of each integer from m to n (subtrahends 1 to 10)
4. Print division table of each integer from m to n (divisors 1 to 10)
5. Exit
Enter your choice (1-5): 1
```

```
Session Actions Edit View Help

(kali@kali)-[~/HDOC_Day21]
$ ls
day21.c
$ gcc day21.c -o day21
$ ./day21
Enter first positive integer (m): 2
Enter second positive integer (n, where n ≥ m): 3

===== MENU (m = 2, n = 3) =====
1. Print first 10 multiples of each integer from m to n
2. Print addition table of each integer from m to n (addends 1 to 10)
3. Print subtraction table of each integer from m to n (subtrahends 1 to 10)
4. Print division table of each integer from m to n (divisors 1 to 10)
5. Exit
Enter your choice (1-5): 1

--- First 10 Multiples of 2 ---
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

--- First 10 Multiples of 3 ---
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

===== MENU (m = 2, n = 3) =====
1. Print first 10 multiples of each integer from m to n
2. Print addition table of each integer from m to n (addends 1 to 10)
3. Print subtraction table of each integer from m to n (subtrahends 1 to 10)
4. Print division table of each integer from m to n (divisors 1 to 10)
5. Exit
Enter your choice (1-5): 4

--- Division Table for 2 (Divisors 1 to 10) ---
2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
2 / 5 = 0.40
2 / 6 = 0.33
2 / 7 = 0.29
2 / 8 = 0.25
2 / 9 = 0.22
2 / 10 = 0.20

--- Division Table for 3 (Divisors 1 to 10) ---
3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
3 / 4 = 0.75
3 / 5 = 0.60
3 / 6 = 0.50
3 / 7 = 0.43
3 / 8 = 0.38
3 / 9 = 0.33
3 / 10 = 0.30

===== MENU (m = 2, n = 3) =====
1. Print first 10 multiples of each integer from m to n
2. Print addition table of each integer from m to n (addends 1 to 10)
3. Print subtraction table of each integer from m to n (subtrahends 1 to 10)
4. Print division table of each integer from m to n (divisors 1 to 10)
5. Exit
Enter your choice (1-5): 5
Exiting program ...
```